Week 3 - Capstone Progress

By: Zackary Siddiqi

TripAlert — Project Progress

Flexible flight + accommodation search with budget price alerts

The Real-World Problem

Budget travelers have a budget and a date window — not fixed dates. Flight & hotel sites require exact dates, forcing long periods of manual checking. Not many tools that combine flexible flight + hotel options into one budget-driven alert system.

Software only — a web application. No physical hardware involved; only standard cloud hosting.

Planned Tools & Tech Stack

Frontend

Next.js + Tailwind CSS

Backend

FastAPI / Node + Express

Database

PostgreSQL

Travel Data

Travelpayouts Data API

Hosting

Vercel(Frontend) + Render (Backend)

Platform, Cost & Extra Details

App Platform

Built as a responsive web app — not a native iOS/Android build.

• Runs on Windows/macOS/Linux too — any modern browser, since it's fully web-based.

Project Costs

$0 planned for the semester — every chosen tool has a free tier sufficient for capstone-scale workload:

• Travelpayouts API — free developer tier (flights + hotels)

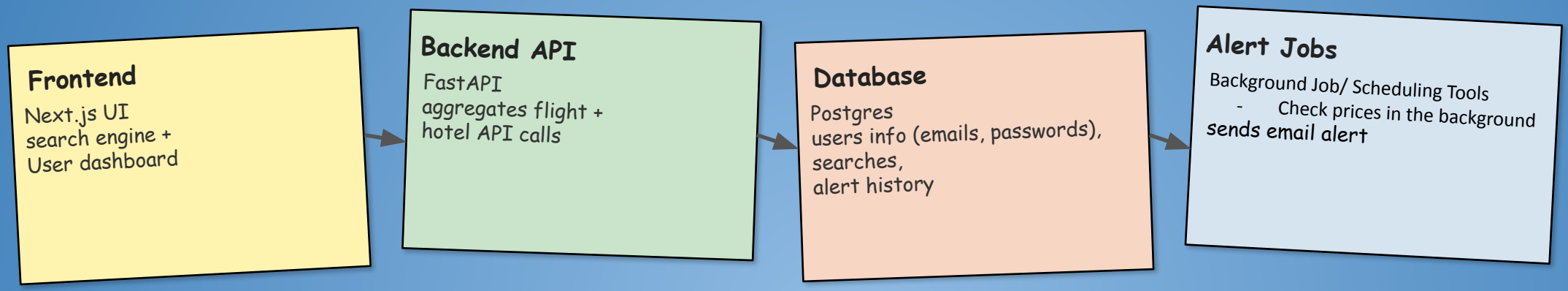
• Vercel + Render — free hosting, DB

Additional Feature

• Stretch feature — **Group Mode:** Allows users to form a group in which they can share the same trip with each other and be able to calculate costs.

Current Progress

System architecture & roadmap sketched out during planning sessions



Current Progress:

